

120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 13: Brownian Motion

Day 91: Ito's Lemma & Euler-Maruyama Method

1 Introduction

Stochastic Differential Equations (SDEs) describe how financial variables evolve over time under randomness.

A general SDE is written as:

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$$

where:

- μ is the drift (deterministic part)
- σ is the volatility (random component)
- W_t is Brownian Motion

2 Ito's Lemma in SDEs

For a function $f(X_t, t)$:

$$df = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW_t$$

This allows transformation of stochastic processes.

3 Euler–Maruyama Method

Since SDEs cannot usually be solved analytically, we approximate them numerically.

The Euler–Maruyama scheme:

$$X_{t+\Delta t} = X_t + \mu(X_t, t)\Delta t + \sigma(X_t, t)\sqrt{\Delta t} \epsilon_t$$

where:

$$\epsilon_t \sim \mathcal{N}(0, 1)$$

4 Example: Geometric Brownian Motion

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Euler approximation:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} \epsilon_t$$

5 Interpretation

- Drift term controls trend
- Diffusion term introduces randomness
- Paths evolve unpredictably over time

6 Financial Relevance

SDEs are used to model:

- stock prices
- interest rates
- volatility

They form the foundation of:

- Black-Scholes model
- stochastic volatility models
- quantitative trading strategies

7 Conclusion

Euler–Maruyama provides a practical way to simulate SDEs.

Combined with Ito’s Lemma, it enables the modeling and transformation of stochastic systems in finance.